Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The primitive of $|x|$, when $x < 0$ is

(A) $\frac{1}{2}x^2 + C$

(B) $-\frac{1}{2}x^2 + C$

(C) $x + C$

(D) $-\mathbf{x} + \mathbf{C}$

Q.2

$\int \frac{\tan(\ln x)\tan\left(\ln \frac{x}{2}\right)\tan(\ln 2)}{x}dx =$

(A) $\ln\left(\frac{\sec(\ln x)}{\sec\left(\ln \frac{x}{2}\right)}\right) + C$

(B) $\ln(\sec \ln x) + C$

(C) $\ln\left(\sec \ln\left(\frac{x}{2}\right) x^{\tan(\ln x)}\right) + C$

(D) $\ln\left(\frac{\sec(\ln \mathrm{x})}{\sec\left(\ln \frac{\mathrm{x}}{2}\right)\mathrm{x}^{\tan(\ln 2)}}\right) + \mathrm{C}$

Q.3

$\int \frac{dx}{\cos x + \sqrt{3}\sin x}$ equals

(A) $\log \tan\left(\frac{x}{2} + \frac{\pi}{12}\right) + C$

(B) $\log \tan\left(\frac{x}{2} - \frac{\pi}{12}\right) + C$

(C) $\frac{1}{2}\log \tan\left(\frac{x}{2} + \frac{\pi}{12}\right) + C$

(D) $\frac{1}{2}\log \tan\left(\frac{x}{2} - \frac{\pi}{12}\right) + C$

Q.4

If $I = \int \frac{1}{2p}\sqrt{\frac{p-1}{p+1}}dp = f(p) + c$, then $f(p)$ is equal to :

(A) $\frac{1}{2}\ell \mathrm{n}\left[\mathrm{p} - \sqrt{\mathrm{p}^2 - 1}\right]$

(B) $\frac{1}{2}\cos^{-1}\mathrm{p} + \frac{1}{2}\sec^{-1}\mathrm{p}$

@IITJEE_Advanced

(C) $\ell n\sqrt{\mathrm{p}+\sqrt{\mathrm{p}^2-1}}-\frac{1}{2}\sec^{-1}\mathrm{p}$

(D) none of the above.

Q.5

If $\int \log_{\mathrm{e}}(\sqrt{1-\mathrm{x}}+\sqrt{1+\mathrm{x}})\mathrm{dx}$
$= x\log_{\mathrm{e}}(\sqrt{1-x}+\sqrt{1+x})+g(x)+C$. Then $g(x)=$

(A) $x-\sin^{-1}x$

(B) $\sin^{-1}x-x$

(C) $x+\sin^{-1}x$

(D) $\sin^{-1}x-x^2$

Q.6

If $\int x\log\left(1+\frac{1}{x}\right)dx$
$=\mathrm{f(x)}\log(\mathrm{x}+1)+\mathrm{g(x)x}^2+\mathrm{Lx}+\mathrm{C}$, then

(A) $f(x)=\frac{1}{2}x^2$

(B) $g(x)=\log x$

(C) $L=1$

(D) none of these

Q.7

$\int x^{51}\left(\tan^{-1}x+\cot^{-1}x\right)dx$

(A) $\frac{x^{52}}{52}\left(\tan^{-1}x+\cot^{-1}x\right)+c$

(B) $\frac{x^{52}}{52}\left(\tan^{-1}x-\cot^{-1}x\right)+c$

(C) $\frac{\pi x^{52}}{104}+\frac{\pi}{2}+c$

(D) $\frac{x^{52}}{52}+\frac{\pi}{2}+c$

Q.8

$\int \mathrm{e}^{3\log x}\left(x^4+1\right)^{-1}dx$ is equal to

(A) $\log\left(x^4+1\right)+C$

(B) $\frac{1}{4}\log\left(x^4+1\right)+C$

(C) $-\log\left(x^4+1\right)+C$

(D) None of these

@IITJEE_Advanced

Q.9

$\int \tan^3 2x \sec 2x dx =$

(A) $\sec^3 2x + 3\sec 2x$

(B) $\frac{1}{6}\left[\sec^3 2x - 3\sec 2x\right.$

(C) $\sec^3 2x - 3\sec 2x$

(D) None of these

Q.10

$\int \cos\sqrt{x} dx =$

(A) $2[\sqrt{x}\sin\sqrt{x} + \cos\sqrt{x}]$

(B) $\sqrt{x}\sin\sqrt{x} - \cos\sqrt{x}$

(C) $2[\sqrt{x}\sin\sqrt{x} - \cos\sqrt{x}]$

(D) None of these

Q.11

$\int \frac{\sqrt{5+x^2}}{x^4} dx$ is equal to

(A) $\frac{1}{15}\left(1+\frac{5}{x^2}\right)^{\frac{3}{2}} + C$

(B) $\frac{-1}{15}\left(1+\frac{1}{x^2}\right)^{\frac{3}{2}} + C$

(C) $\frac{-1}{15}\left(1+\frac{5}{x^2}\right)^{\frac{3}{2}} + C$

(D) $\frac{1}{15}\left(1+\frac{1}{x^2}\right)^{\frac{3}{2}} + C$

Q.12

$\int \frac{\sqrt{x}}{1+\sqrt[4]{x^3}} dx$ is equal to

(A) $\frac{4}{3}\left[1 + x^{3/4} + \log\left(1+x^{3/4}\right)\right] + C$

(B) $\frac{4}{3}\left[1 + x^{3/4} - \log\left(1+x^{3/4}\right)\right] + C$

(C) $\frac{4}{3}\left[1 - x^{3/4} + \log\left(1+x^{3/4}\right)\right] + C$

(D) None of these

Q.13

$\int \frac{\left(x^2-1\right)}{x\sqrt{x^4+3x^2+1}} dx$ is equal to

@IITJEE_Advanced

(A) $\log\left|\mathrm{x}+\frac{1}{\mathrm{x}}+\sqrt{\mathrm{x}^2+\frac{1}{\mathrm{x}^2}+3}\right|+\mathrm{C}$

(B) $\log\left|\mathrm{x}-\frac{1}{\mathrm{x}}+\sqrt{\mathrm{x}^2+\frac{1}{\mathrm{x}^2}-3}\right|+\mathrm{C}$

(C) $\log\left|x+\sqrt{x^2+3}\right|+C$

(D) None of these

Q.14

$\int \frac{(1+x)e^x}{\cot(xe^x)}dx$ is equal to

(A) $\log|\cos(xe^x)|+C$

(B) $\log|\cot(xe^x)|+C$

(C) $\log|\sec(xe^{-x})|+C$

(D) $\log|\sec(xe^x)|+C$

Q.15

$\int \left(x+\frac{1}{x}\right)^{n+5}\left(\frac{x^2-1}{x^2}\right)dx$ is equal to

(A) $\frac{\left(x+\frac{1}{x}\right)^{n+6}}{n+6}+c$

(B) $\left[\frac{x^2+1}{x^2}\right]^{n+6}(n+6)+c$

(C) $\left[\frac{x}{x^2+1}\right]^{n+6}(n+6)+c$

(D) none of these

Q.16

What is $\int \sec^n x\tan x dx$ equal to ?

(A) $\frac{\sec^n x}{n}+c$

(B) $\frac{\sec^{n-1} x}{n-1}+c$

(C) $\frac{\tan^n x}{n}+c$

(D) $\frac{\tan^{n-1} x}{n-1}+c$

Where ' c ' is a constant of integration.

Q.17

If $\int \sin^3 x\cos^5 x dx$
$=A\sin^4 x+B\sin^6 x+C\sin^8 x+D$ Then

(A) $A=\frac{1}{4},\quad B=-\frac{1}{3}, C=\frac{1}{8}, D\in R$

@IITJEE_Advanced

(B) $\mathrm{A}=\frac{1}{8}, \mathrm{B}=\frac{1}{4}, \mathrm{C}=\frac{1}{3}, \mathrm{D} \in \mathrm{R}$

(C) $A=0, \quad B=-\frac{1}{6}, C=\frac{1}{8}, D \in R$

(D) None of these

Q.18

Evaluate: $\int \frac{1}{1+3 \sin^2 x+8 \cos^2 x} dx$

(A) $\frac{1}{6} \tan^{-1}(2 \tan x)+C$

(B) $\tan^{-1}(2 \tan x)+C$

(C) $\frac{1}{6} \tan^{-1}\left(\frac{2 \tan x}{3}\right)+C$

(D) None of these

Q.19

$\phi(x)=\int \cot^4 x dx+\frac{1}{3} \cot^3 x-\cot x$
and $\phi\left(\frac{\pi}{2}\right)=\frac{\pi}{2}$ then $\phi(\mathrm{x})$ is

(A) $\pi-x$

(B) $x-\pi$

(C) $\pi/2-x$

(D) x

Q.20

The value of $\int \mathrm{e}^{\tan^{-1} \mathrm{x}} \frac{\left(1+\mathrm{x}+\mathrm{x}^2\right)}{1+\mathrm{x}^2} \mathrm{dx}$ is

(A) $xe^{\tan^{-1}} x+c$

(B) $\tan^{-1} x+C$

(C) $\mathrm{e}^{\tan^{-1} \mathrm{x}}+2 \mathrm{x}+\mathrm{C}$

(D) none of these.

Answers & Solutions

Q.1 Answer:

$-\frac{1}{2}x^2+C$

Solution:

$\frac{d}{dx}\left(-\frac{1}{2}x^2+C\right)=-x$

Q.2 Answer:

@IITJEE_Advanced

$$\ln\left(\frac{\sec(\ln x)}{\sec\left(\ln\frac{x}{2}\right)x^{\tan(\ln 2)}}\right)+C$$

Solution:

$$\ln x=\ln\left(\frac{x}{2}\right)+\ln 2$$
$$\Rightarrow \tan(\ln x)=\frac{\tan(\ln x/2)+\tan(\ln 2)}{1-\tan(\ln x/2)\tan(\ln 2)}$$
$$\Rightarrow \tan(\ln x)\tan\left(\ln\frac{x}{2}\right)\tan(\ln 2)=\tan(\ln x)-\tan\left(\ln\frac{x}{2}\right)-\tan(\ln 2)$$
$$\therefore \quad I=\int\frac{\tan(\ln x)}{x}dx-\int\frac{\tan(\ln x/2)}{x}dx-\int\frac{\tan(\ln 2)}{x}dx$$
$$=\ln\sec(\ln x)-\ln\sec\left(\ln\frac{x}{2}\right)-\tan(\ln 2)\ln x+C$$
$$=\ln\left[\frac{\sec(\ln x)}{\sec\left(\ln\frac{x}{2}\right)}\right]-\ln x^{\tan(\ln 2)}+C$$
$$=\ln\left[\frac{\sec(\ln x)}{\sec\left(\ln\frac{x}{2}\right)x^{\text{son}(\ln 2)}}\right]+C$$

Q.3 Answer:

$\frac{1}{2}\log\tan\left(\frac{x}{2}+\frac{\pi}{12}\right)+C$

Solution:

$$I=\int\frac{dx}{\cos x+\sqrt{3}\sin x}$$
$$\Rightarrow I=\int\frac{dx}{2\left[\frac{1}{2}\cos x+\frac{\sqrt{3}}{2}\sin x\right]}$$
$$=\frac{1}{2}\int\frac{\mathrm{dx}}{\left[\sin\frac{\pi}{6}\cos\mathrm{x}+\cos\frac{\pi}{6}\sin\mathrm{x}\right]}$$
$$=\frac{1}{2}\cdot\int\frac{\mathrm{dx}}{\sin\left(\mathrm{x}+\frac{\pi}{6}\right)}$$
$$\Rightarrow \quad \mathrm{I}=\frac{1}{2}\cdot\int\mathrm{cosec}\left(\mathrm{x}+\frac{\pi}{6}\right)\mathrm{dx}$$
$$\because\int\mathrm{cosec}\,x dx=\log|(\tan x/2)|+C$$
$$\therefore \mathrm{I}=\frac{1}{2}\cdot\log\tan\left(\frac{\mathrm{x}}{2}+\frac{\pi}{12}\right)+\mathrm{C}$$

Q.4 Answer:

$\ell n\sqrt{\mathrm{p}+\sqrt{\mathrm{p}^2-1}}-\frac{1}{2}\sec^{-1}\mathrm{p}$

Solution:

$$\mathrm{I}=\int\frac{1}{2\mathrm{p}}\sqrt{\frac{\mathrm{p}-1}{\mathrm{p}+1}}\mathrm{dp}$$
$$=\frac{1}{2}\int\frac{\mathrm{p}-1}{\mathrm{p}\sqrt{(\mathrm{p}+1)(\mathrm{p}-1)}}\mathrm{dp}$$
$$=\frac{1}{2}\int\frac{\mathrm{pdp}}{\mathrm{p}\sqrt{\mathrm{p}^2-1}}-\frac{1}{2}\int\frac{\mathrm{dp}}{\mathrm{p}\sqrt{\mathrm{p}^2-1}}$$
$$=\frac{1}{2}\int\frac{\mathrm{dp}}{\sqrt{\mathrm{p}^2-1}}-\frac{1}{2}\int\frac{\mathrm{dp}}{\mathrm{p}\sqrt{\mathrm{p}^2-1}}$$
$$=\frac{1}{2}\log_{\mathrm{e}}\left(\mathrm{p}+\sqrt{\mathrm{p}^2-1}\right)-\frac{1}{2}\sec^{-1}\mathrm{p}$$
$$\Rightarrow f(p)=\log\sqrt{p+\sqrt{p^2-1}}-\frac{1}{2}\sec^{-1}p$$

Q.5 Answer:

$\sin^{-1}x-x$

@IITJEE_Advanced

Solution:

$I = \int \log_e(\sqrt{1-x} + \sqrt{1+x}) \cdot 1 dx$

Integrating by parts taking 1 as the second function.

$I = \log(\sqrt{1-x} + \sqrt{1+x})x - \int \frac{1}{\sqrt{1-x}+\sqrt{1+x}}$
$\left[-\frac{1}{2\sqrt{1-x}} + \frac{1}{2\sqrt{1+x}}\right](x)dx$
$= x\log(\sqrt{1-x} + \sqrt{1+x})$
$-\frac{1}{2}\int \frac{\sqrt{1-x}-\sqrt{1+x}}{\sqrt{1-x}+\sqrt{1+x}} \cdot \frac{1}{\sqrt{1-x^2}} \cdot x dx$
$= x\log(\sqrt{1-x} + \sqrt{1+x})$
$-\frac{1}{2}\int \frac{(1-x)+(1+x)-2\sqrt{1-x^2}}{(1-x)-(1+x)} \cdot \frac{1}{\sqrt{1-x^2}} \cdot x dx$
$= x\log(\sqrt{1-x} + \sqrt{1+x}) - \frac{1}{2}\int \frac{\sqrt{1-x^2}-1}{\sqrt{1-x^2}} dx$
$= x\log(\sqrt{1-x} + \sqrt{1+x}) - \frac{1}{2}\left[\int 1 dx - \int \frac{1}{\sqrt{1-x^2}} dx\right]$
$= x\log(\sqrt{1-x} + \sqrt{1+x}) + \frac{1}{2}\left[\sin^{-1} x - x\right] + C$
$\therefore \mathrm{f(x)} = \mathrm{x}, \mathrm{g(x)} = \sin^{-1}\mathrm{x} - \mathrm{x}$

Answer is g(x)/2

Q.6 Answer:

none of these

Solution:

$\int x\log\left(1+\frac{1}{x}\right) dx$
$= \log\left(1+\frac{1}{x}\right) \cdot \frac{x^2}{2} - \int \frac{x}{x+1} \cdot \left(-\frac{1}{x^2}\right) \cdot \frac{x^2}{2} dx$
$= \frac{x^2}{2}\log\left(\frac{x+1}{x}\right) \cdot \frac{x^2}{2} + \frac{1}{2}\int \frac{x+1-1}{x+1} dx$
$= \frac{x^2}{2}\log\left(\frac{x+1}{x}\right) + \frac{1}{2}x - \frac{1}{2}\log(x+1) + c$
$= \left(\frac{x^2-1}{2}\right)\log(x+1) - \frac{x^2}{2}\log x + \frac{1}{2}x + c$

Q.7 Answer:

$\frac{x^{52}}{52}\left(\tan^{-1} x + \cot^{-1} x\right) + c$

Solution:

$\int x^{51}\left(\tan^{-1} x + \cot^{-1} x\right) dx$
$= \int \mathrm{x}^{51} \cdot \frac{\pi}{2} \mathrm{dx} \quad \left\{\because \tan^{-1}\mathrm{x} + \cot^{-1}\mathrm{x} = \frac{\pi}{2}\right\}$
$= \frac{\pi x^{52}}{104} + c = \frac{x^{52}}{52}\left(\tan^{-1} x + \cot^{-1} x\right) + c$

Q.8 Answer:

$\frac{1}{4}\log\left(x^4+1\right) + C$

Solution:

$\int e^{3\log x}\left(x^4+1\right)^{-1} dx = \int e^{\log x^3} \frac{1}{x^4+1} dx$
$= \int \frac{x^3}{x^4+1} dx = \frac{1}{4}\log\left(x^4+1\right) + C$

@IITJEE_Advanced

$\left[\text{ since } e^{\log_e x^3} = x^3\right]$

Q.9 Answer:

$\frac{1}{6}\left[\sec^3 2x - 3\sec 2x\right.$

Solution:

Let $\quad \mathrm{I} = \int \tan^3 2\mathrm{x} \cdot \sec 2\mathrm{x}\mathrm{dx}$

$= \int \tan^2 2x \cdot \sec 2x \cdot \tan 2x dx$

$= \int \left(\sec^2 2x - 1\right) \sec 2x \cdot \tan 2x dx$

Put sec $2\mathrm{x} = \mathrm{t}$

$2 \sec 2x \cdot \tan 2x dx = dt$

Now $\sec 2x \cdot \tan 2x dx = \frac{dt}{2}$

$\mathrm{I} = \frac{1}{2} \int \left(\mathrm{t}^2 - 1\right) \mathrm{dt}$

$= \frac{1}{2}\left[\frac{\mathrm{t}^3}{3} - \mathrm{t}\right] + \mathrm{C}$

$= \frac{1}{2}\left[\frac{\sec^3 2x}{3} - \sec 2x\right] + C = 1/6\left[\sec^3 2x - 3\sec 2x\right] + C$

Q.10 Answer:

$2[\sqrt{x}\sin\sqrt{x} + \cos\sqrt{x}]$

Solution:

$\mathrm{I} = \int \cos\sqrt{\mathrm{x}}\mathrm{dx}$

Put $\sqrt{\mathrm{x}} = \mathrm{t}$ **so that** $\frac{1}{2\sqrt{(\mathrm{x})}}\mathrm{dx} = \mathrm{dt}$

$\therefore I = f(\cos t)2tdt = 2\int t\cos t dt$

Integrating by parts, we get

$I = 2\left[t\sin t - \int 1 \cdot \sin t dt\right]$

$= 2[t\sin t + \cos t]$

$= 2[\sqrt{x}\sin\sqrt{x} + \cos\sqrt{x}]$

Q.11 Answer:

$\frac{-1}{15}\left(1 + \frac{5}{x^2}\right)^{\frac{3}{2}} + C$

Solution:

Let $\mathrm{I} = \int \frac{\sqrt{5+x^2}}{x^4} dx$

put $x = \sqrt{5}\tan\theta$

$\Rightarrow \int \frac{\sqrt{5+x^2}}{x^4} dx = \int \frac{\sqrt{5+5\tan^2\theta}}{(\sqrt{5}\tan\theta)^4}\sqrt{5} \cdot \sec^2\theta d\theta$

$= \int \frac{\sqrt{5(1+\tan^2\theta)}}{(5)^2\tan^4\theta}\sqrt{5} \cdot \sec^2\theta d\theta$

$= \int \frac{\sqrt{5}\sec^2\theta}{25\tan^4\theta}\sqrt{5}\sec^2\theta d\theta$

$= \int \frac{\sqrt{5}\sec\theta\sqrt{5}\sec^2\theta d\theta}{25\tan^4\theta} = \frac{1}{5}\int \frac{\cos\theta}{\sin^4\theta} d\theta$

Put $\sin\theta = t \Rightarrow \cos\theta \mathrm{d}\theta = \mathrm{dt}$

$= \frac{1}{5}\int \frac{dt}{t^4} = \frac{-1}{15}\frac{1}{t^3} + \mathrm{C} = \frac{-1}{15}\left[\frac{1}{\sin^3\theta}\right] + \mathrm{C}$

@IITJEE_Advanced

$= \frac{-1}{15}\left[\operatorname{cosec}^2\theta \cdot \operatorname{cosec}\theta\right] + \mathrm{C}$

$= \frac{-1}{15}\left[\left(1+\cot^2\theta\right)\cdot\sqrt{1+\cot^2\theta}\right] + \mathrm{C}$

$= \frac{-1}{15}\left[\frac{1+\tan^2\theta}{\tan^2\theta} \times \sqrt{\frac{1+\tan^2\theta}{\tan^2\theta}}\right] + \mathrm{C}$

$= -\frac{1}{15}\left[\frac{1+\tan^2\theta}{\tan^2\theta}\right]^{3/2} + \mathrm{C} = -\frac{1}{15}\left[\frac{1+\frac{x^2}{5}}{\frac{x^2}{5}}\right]^{3/2} + \mathrm{C}$

$= \frac{-1}{15}\left[1+\frac{5}{x^2}\right]^{\frac{3}{2}} + \mathrm{C}$

Q.12 Answer:

$\frac{4}{3}\left[1+x^{3/4}-\log\left(1+x^{3/4}\right)\right]+C$

Solution:

Put $x = z^4 \Rightarrow dx = 4z^3dz$

$\therefore \int \frac{\sqrt[2]{x}}{1+\sqrt[4]{x^3}}dx = \int \frac{z^2\cdot 4z^3}{1+z^3}dz = 4\int \frac{z^3\cdot z^2}{z^3+1}dz$

$= \frac{4}{3}\int \frac{(\mathrm{y}-1)}{\mathrm{y}}\mathrm{dy}$

[Putting $z^3+1 = y \Rightarrow z^2dz = \frac{1}{3}dy$

$= \frac{4}{3}(y-\log y)+C$

$= \frac{4}{3}\left[1+x^{3/4}-\log\left(1+x^{3/4}\right)\right]+C$

Q.13 Answer:

$\log\left|\mathrm{x}+\frac{1}{\mathrm{x}}+\sqrt{\mathrm{x}^2+\frac{1}{\mathrm{x}^2}+3}\right|+\mathrm{C}$

Solution:

$\int \frac{\left(x^2-1\right)}{x\sqrt{x^4+3x^2+1}}dx = \int \frac{\left(x^2-1\right)}{x^2\sqrt{x^2+3+\frac{1}{x^2}}}dx$

$= \int \frac{\left(1-\frac{1}{x^2}\right)}{\sqrt{\left(x+\frac{1}{x}\right)^2+1}}dx = \int \frac{dz}{\sqrt{z^2+1}}$

[**Putting** $x+\frac{1}{x} = z \Rightarrow \left(1-\frac{1}{x^2}\right)dx = dz$

$= \log\left|z+\sqrt{z^2+1}\right|+C$

$= \log\left|x+\frac{1}{x}+\sqrt{x^2+\frac{1}{x^2}+3}\right|+C$

Q.14 Answer:

$\log\left|\sec\left(xe^x\right)\right|+C$

Solution:

Let $\mathrm{I} = \int \frac{(1+x)e^x}{\cot(xe^x)}dx$

Put $xe^x = t \Rightarrow \left(xe^x+e^x\right)dx = dt$

$\Rightarrow e^x(x+1)dx = dt$

@IITJEE_Advanced

$\therefore \mathrm{I}=\int \frac{dt}{\cot(t)}=\log|\sec t|+\mathrm{C}$
$=\log|\sec(xe^x)|+C$

Q.15 Answer:

$\frac{\left(x+\frac{1}{x}\right)^{n+6}}{n+6}+c$

Solution:

$\mathrm{I}=\int\left(\mathrm{x}+\frac{1}{\mathrm{x}}\right)^{\mathrm{n}+5}\left(\frac{\mathrm{x}^2-1}{\mathrm{x}^2}\right)\mathrm{dx}$
Put $x+\frac{1}{x}=t\Rightarrow\left(1-\frac{1}{x^2}\right)dx=dt$
$\Rightarrow\left(\frac{\mathrm{x}^2-1}{\mathrm{x}^2}\right)\mathrm{dx}=\mathrm{dt}$
$\therefore I=\int t^{n+5}dt=\frac{t^{n+6}}{n+6}+c=\frac{\left(x+\frac{1}{x}\right)^{n+6}}{n+6}+c$

Q.16 Answer:

$\frac{\sec^n x}{n}+c$

Solution:

Let $\mathrm{I}=\int\sec^n x\tan x dx.$
Put, $\sec x=t\Rightarrow\sec x\tan x dx=dt$
$\therefore I=\int t^n\cdot\frac{dt}{t}$
$\int t^{n-1}dt=\frac{t^n}{n}+c=\frac{\sec^n x}{n}+c$
where ' C ' is a constant of integration

Q.17 Answer:

$A=\frac{1}{4},\quad B=-\frac{1}{3},C=\frac{1}{8},D\in R$

Solution:

$I=\int\sin^3 x\cdot\cos^5 x dx$
Put $\sin x=t\Rightarrow\cos x dx=dt$
$I=\int\sin^3 x\cdot\cos^4 x\cdot\cos x dx=\int t^3\left(1-t^2\right)^2dt$
$=\int\left(t^3-2t^5+t^7\right)dt=\frac{1}{4}t^4-\frac{2}{6}t^6+\frac{1}{8}t^8+D$
$=\frac{1}{4}\sin^4\mathrm{x}-\frac{1}{3}\sin^6\mathrm{x}+\frac{1}{8}\sin^8\mathrm{x}+\mathrm{D}$

Q.18 Answer:

$\frac{1}{6}\tan^{-1}\left(\frac{2\tan x}{3}\right)+C$

Solution:

$\mathrm{I}=\int\frac{1}{1+3\sin^2\mathrm{x}+8\cos^2\mathrm{x}}\mathrm{dx}$
Dividing the numerator and denominator by $\cos^2\mathrm{x}$**, we get**
$I=\int\frac{\sec^2 x}{\sec^2 x+3\tan^2 x+8}dx$
$\Rightarrow I=\int\frac{\sec^2 x}{1+\tan^2 x+3\tan^2 x+8}dx=\int\frac{\sec^2 x}{4\tan^2 x+9}dx$

@IITJEE_Advanced

Putting $\tan x = t \Rightarrow \sec^2 x dx = dt$, we get

$$I = \int \frac{dt}{4t^2+9} = \frac{1}{4}\int \frac{dt}{t^2+(3/2)^2} = \frac{1}{4} \times \frac{1}{3/2} \tan^{-1}\left(\frac{t}{3/2}\right) + C$$

$$\Rightarrow \quad I = \frac{1}{6}\tan^{-1}\left(\frac{2t}{3}\right) + C = \frac{1}{6}\tan^{-1}\left(\frac{2\tan x}{3}\right) + C$$

Q.19 Answer:

x

Solution:

$$\int \cot^4 x dx = \int \cot^2 x \cdot \left(\operatorname{cosec}^2 x - 1\right) dx$$

$$= \int \cot^2 x \operatorname{cosec}^2 x dx - \int \left(\operatorname{cosec}^2 x - 1\right) dx$$

$$= -\frac{1}{3}\cot^3 x + \cot x + x + c$$

$$\therefore \phi(x) = -\frac{1}{3}\cot^3 x + \cot x + x + c + \frac{1}{3}\cot^3 x - \cot x$$

$$= x + c$$

$$\therefore \phi\left(\frac{\pi}{2}\right) = \frac{\pi}{2} + c, \quad \therefore c = 0, \therefore \phi(x) = x$$

Q.20 Answer:

$xe^{\tan^{-1} x} + c$

Solution:

$$\tan^{-1} x = t \Rightarrow x = \tan t$$

$$dx \cdot \frac{1}{1+x^2} = dt$$

$$= \int e^t (\tan t + \sec^2 t)\, dt \quad \left[\because \int e^x \left(f(x) + f'(x) \cdot dx = e^x f(x) + c\right]\right.$$

$$I = \int e^t \left(1 + \tan t + \tan^2 t\right) dt$$

$$= e^t \tan t + c$$

$$\tan^{-1} x \tan\left(\tan^{-1} x\right) + c$$

$$I = x e^{\tan^{-1} x} + c$$

@IITJEE_Advanced